Continuous Delivery Architect (CDA) Examination

Sample Paper

Multiple Choice

Instructions

1. All 40 questions should be attempted.
2. All answers are to be marked on the answer sheet provided.
3. Please use a pencil and NOT ink to mark your answers on the answer sheet provided. There is only one correct answer per question.
4. You have 1 ½ hours to complete this paper.
5. You must achieve 26 or more out of a possible 40 marks (65%) to pass this exam.

Sam

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

1. **An organization wants to increase the flow of software delivery by taking a Continuous Delivery approach. Which of the following will be important to their success?**

    a. Agile software development

    b. A monolithic architecture

    c. A straightforward and repeatable deployment process

    d. Open source tools connected through APIs

2. **Which of the following describes Continuous Deployment?**

    a. End-to-end pipeline

    b. Final stage of the pipeline responsible for packaging, releasing and distributing artifacts to production for customer use

    c. Pipeline stage where final packaging and acceptance testing is performed

    d. An automated approach to release engineering

3. **An organization wants to improve and expedite the interaction between Operations, Security and Development teams. How would embracing the Second Way from the Phoenix Project help to achieve this goal?**

    a. They would try to create a culture that fosters risk taking.

    b. The would understand that repetition and practice is the prerequisite to mastery

    c. They would use tools such as Chat Ops to shorten feedback loops

    d. They would use automation to increase the flow of work

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

4. **Which of the following BEST describes the key principles of designing for Continuous Delivery?**

    a. Refactoring code into modules, use Agile practices, pre-flight test all changes

    b. Automate quality testing of modules and collaborate with Operations

    c. Build quality in, work in small batches, embrace automation, continuous improvements

    d. Advocate best practices, promote design improvements, participate in architecture reviews

5. **Which of the following are "illities" that most matter to Continuous Delivery pipelines?**

    a. Reliability, Usability, Maintainability

    b. Test-ability, Configurability, Monitor-ability

    c. Evolve-ability, Security, Usability

    d. Reproduce-ability, Deploy-ability, Quality

6. **An organization is implementing Continuous Integration practices. Which of the following would improve the ability of developers to rebase or shelve changes before pushing to other users?**

    a. Microservices

    b. Agile software development

    c. Artifact repository

    d. A distributed version control system

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

7. **Which of the following is a best practice for a containers test strategy?**

    a. Create an integration test container separate from the application code containers

    b. Keep testing tools and artifacts in the same container as the application code

    c. Perform Blue-Green testing before deployment

    d. Verify backwards and forwards compatibility policies for micro-services

8. **Why is it important to manage test versions?**

    a. Tests are as important as the code because the tests are the validation of the code

    b. Developers would need models for test driven development

    c. Infrastructure teams would not need to run their own tests

    d. The test versions can be reused across platforms

9. **Which of the following BEST describes the recommended steps for Gradual Feature-Flag Rollouts?**

    a. Rollout changes to the green nodes, monitor performance, if failures occur revert to blue nodes, otherwise rollout to the remaining users

    b. Rollout changes to a select set of users, if no failures rollout to the rest of the users

    c. Turn on the feature flags for all users, monitor performance and turn off the flags for any users that experience failures

    d. Gradually increase rollout of changes to user nodes, revert if failures occur, measure improvements, and continue until rollout is completed

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

10. Which of the following correctly captures Jez Humble's definition of Continuous Delivery?

a. A set of practices that emphasize the **collaboration** and **communication** of both software developers and information technology (IT) professionals while automating the process of software delivery and infrastructure changes

b. A software engineering approach in which teams produce software in short cycles, ensuring that the software can be reliably released at any time

c. The ability to get changes—features, configuration changes, bug fixes, experiments—into production or into the hands of users safely and quickly in a sustainable way

d. A journey toward sound engineering and delivery principles, and it starts with breaking down silos and mapping out the entire delivery pipeline so that it can be optimized

11. An organization has some monitoring capability in place but wants to move towards Continuous Monitoring. Why is Continuous Monitoring a key pillar of Continuous Delivery?

a. Continuous monitoring provides visibility of the health of all things that are important to the operation and performance of the continuous delivery pipeline

b. The health of all seven pillars affects the performance of continuous delivery

c. Continuous monitoring applies to the other six pillars and itself

d. All of the above

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

12. XYZ Corporation is building their deployment pipeline toolchain. What should be their first step?

a. Assess their current infrastructure and tools

b. Model their value stream

c. Automate their unit tests and code analysis

d. Create a reference architecture of available open source tools

13. Which of the following concepts seeks to instill a "security as code" culture?

a. DevSecOps

b. Security assurance

c. Rugged DevOps

d. Conway's Law

14. Which of the following is NOT a best practice for securing containers?

a. Use containers on workloads of similar trust levels; use hypervisors or physical separation for strongest security levels

b. Do not use whitelists for containers because it would be redundant to do so

c. Use image signing to provide a chain of custody

d. Log admin access for evidence of a breach, and compliance audits

15. Which of the following is NOT a good example of Continuous Integration monitoring metrics?

a. Unit test pass rate

b. Build success rate

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

c. Integration test coverage

d. Application performance test results

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

16. Which of the following is a best practice for converting a Monolith to Microservice architecture?

a. Refactor the monolith in a feature branch then merge the completed microservices based system into mainline after testing is completed

b. Start with low risk code areas

c. Look for *seams* – areas of code that are independent, focused around a single business capability

d. Start by splitting business logic dependencies conversions then tackle data dependencies

17. Which of the following are important team agreements associated with Continuous Integration?

a. Everyone agrees to check in changes to mainline when the feature is completed and is to respond to bugs reported by QA

b. Everyone agrees to thoroughly test all changes prior to check-in changes to mainline

c. Everyone agrees to report known problems with check-ins and to schedule fixes according to priority levels

d. agrees to check in small incremental changes frequently to the mainline and that the highest priority task on the project is to fix any change that breaks the application

18. Which of the following is NOT a typical role for an artifact repository?

a. Store binaries for specific builds

b. Store metadata for release candidates

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

c. Store source code changes to be reused to make builds for maintenance releases

d. Store test reports for release candidate builds

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

19. An organization is planning for Continuous Testing. Which best practice should they build into their plans?

a. Since Continuous Testing is performed continuously on incremental changesthere is no need for a separate test plans

b. Test plans should be reviewed by multiple resources to ensure that the plans have sufficient test coverage to meet multiple requirements including infrastructure and tools

c. They can limit their plans to defining specific tests for each incremental change

d. Test plans can be written after the tests are completed to provide a record for compliance audit purposes

20. Conway's Law is significant to Continuous Delivery because

a. It recognizes that an organization will be constrained to produce designs which are copies of their communication structures

b. It will promote the fact that quality is everyone's responsibility

c. It recognizes that a process is only as successful as its weakest link

d. It helps organizations understand that value is only defined from the perspective of the end customer

21. Why is a collaborative culture critical for Continuous Delivery?

a. Culture is the domain of DevOps and is outside the scope of Continuous Delivery systems

b. Technology alone will not provide effective end-to-end pipelines if enterprise leaders, middle managers, development, Quality Assurance, infrastructure and operational teams do not cooperate with each other

c. Continuous Delivery is affected by culture; however a

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

collaborative culture is not considered essential for continuous delivery

d. As part of a Continuous Delivery Pipeline Collaborative Culture is handled by collaborative tools such as chat systems

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

22. An organization is struggling with creating a culture that will support Continuous Delivery and DevOps. They have designated and trained an individual to be their first Continuous Delivery Architect (CDA). What can the CDA do to instill a collaborative culture in their organization?

a. State their views as questions instead of assertions

b. Recognize the contributions of others

c. Move out of their private zone to connect with others

d. All of the above

23. Which of the following is NOT a core principle for Continuous Deployment?

a. Automate the build, deploy, test, and release process

b. Deploy frequent, small releases

c. Get changes into production or into the hands of users safely and quickly in a sustainable way

d. Deliver feedback as quickly as possible

24. Which of the following is a benefit of feature toggles?

a. Allows selective deployments to predetermined “trial” customers

b. Enables developers to evaluate features and alternatives designs

c. Integrates well with a Blue Green Strategy

d. Creates a collaborative culture

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

25. ABC Corporation is aligning Continuous Integration with their software development practices. However, they are having a difficult time optimizing their Continuous Integration workflows and their Continuous Integration processes are not accelerating. What might they be doing wrong?

a. They are stopping to fix any problems that fails integration before proceeding

b. They are pretesting and prechecking integration deliverables

c. They are running the most important tests early

d. They are using analytics to detect threshold exceptions

26. What are the cultural requirements for effective DevOps including Continuous Delivery?

a. n, collaboration, tooling, scale

b. Collaboration, affinity, tooling, scale

c. Collaboration, affinity, tooling, leadership

d. Collaboration, acceptance, tooling, scale

27. Why is an immutable infrastructure desirable for Continuous Delivery?

a. Avoids image and configuration drift

b. More fail-safe than configuration management

c. Easy rollback

d. All of the above

28. Why should organizations move towards Continuous Testing?

a. It reduces costs and time to market

b. It improves innovation

c. It eliminates the need for test plans

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

d. It improves software quality

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

29. Which of the following describes best practices for monitoring databases?

a. There is no need to additional monitoring of databases. Most database tools have internal capabilities for monitoring

b. Establish alerts for database threshold violations

c. Monitor the size and growth of the database

d. Trend resource consumption trends and response times

30. Which of the following IS NOT a characteristic of a Continuous Delivery pipeline?

a. It is a series of processes which are performed on product changes in stages

b. Each stage processes the artifacts resulting from the prior stage

c. The only changes that go into the pipeline are new versions of code, data or images for applications

d. It is the same as a DevOps toolchain

31. Which of the following BEST describes the consequences of not doing Continuous Delivery properly?

a. Scaling issues, quality problems, pipeline failures, reverts, process delays, schedule delays, cost overruns, audit failures

b. Security problems, quality problems, pipeline failures, reverts, process delays, schedule delays, cost overruns, audit failures

c. Attrition, customer dissatisfaction, quality problems,

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

pipeline failures, reverts, process delays, schedule delays, cost overruns, audit failures

d. Breakdown of Conway's Law, numerous process constraints, pipeline failures, reverts, process delays, cost overruns, audit failures

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

32. An organization has established a rapid response team to address issues and problems across the value stream. How can the Continuous Delivery Architect (CDA) support this team?

a. By defining and maintaining roles, responsibilities, service level agreements and monitoring techniques for quick problem identification and response

b. By ensuring that the artifact repository is available for remediation and roll-back procedures

c. By ensuring that commits are gated so that they cannot progress without passing certain tests

d. By implementing ChatOps tools to help with faster collaboration amount teams

33. A manufacturing company is attempting to implement Continuous Testing. What strategies should they consider when developing their automation strategy?

a. New automated tests should be created at the same time application code changes are created

b. Automated tests should match the specific content being changed

c. Test results should be produced automatically

d. Automated tests should only be run on lower risk application code changes

34. Which of the following is a best practice for test failure remediation?

a. Dev and test team members collaborate to resolve defects

b. Always stop the pipeline when a test fails, diagnose and fix the problem and then continue

c. Test failures are not allowed to interrupt the pipeline. The

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

failure is reported and the pipeline continues

d. Any test that fails is assumed to be a problem with the test itself. The test is isolated for review to determine how best to fix the test

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

35. Which of the following is a NOT typical step in Continuous Integration workflows?

a. Verify merge gate threshold are satisfied

b. Merge changes with the mainline repository

c. Deploy the build

d. Release the build resources back to the pool

36. How does applying 12-factor application design practices make implementing Continuous Delivery easier?

a. They support deployment in cloud environments and requires minimal administration

b. They support automatic creation of APIs between applications and infrastructure

c. They support portability between execution environments

d. They minimize divergence between dev and production

37. A large enterprise is attempting to re-architect their legacy, monolithic applications into microservices. How would the company benefit from this effort?

a. Improved productivity

b. Better resource utilization

c. Easier to package / deploy /change

d. Better automation capabilities

38. A dark launching release strategy is particularly valuable for

a. Continuous Delivery

b. Continuous Deployment

c. Security testing

d. Continuous Integration

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

39. What would be a key benefit of frequent, smaller, incremental integrations?

a. The root cause of integration problems can be isolated much faster when the changes are integrated incrementally

b. It will become easier and more necessary to apply Agile software development practices

c. Integration problems can be identified during any stage of the Continuous Delivery Pipeline as part of Continuous Testing

d. Frequent integrations can help infrastructure teams identify additional capacity requirements

40. A Continuous Delivery Architect (CDA) can be a strong influencer on DevOps culture. What can a CDA do to help transform a traditional culture into one that can be successful as a DevOps environment?

a. Lead architecture reviews and create prescriptive architecture checklists

b. Advocate program investments and best practices for collaborative tools

c. Research the latest trends in open source DevOps and Continuous Delivery tools

d. Mentor cross-functional team members and create training programs to help them gain best practices and tool expertise

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

Continuous Delivery Architect
Sample Exam Answer Key

Question	Answer	Topic Area	Slide Reference
1	C	1	16
2	B	1	19
3	C	2	58
4	C	3	78
5	B	3	90
6	D	4	109
7	A	5	159
8	A	5	151
9	D	6	186
10	C	1	17
11	D	7	193
12	B	8	222
13	A	9	243
14	B	9	257
15	D	7	201
16	C	3	94
17	D	4	106
18	C	4	110
19	B	5	146
20	A	2	49
21	B	2	25
22	D	2	57
23	C	6	173
24	A	6	185
25	A	4	127

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12:

26	B	1	51
27	D	8	228
28	C	5	137
29	A	7	200
30	D	1	28
31	B	1	40
32	A	2	62
33	D	5	148
34	A	5	163
35	C	4	114
36	B	3	82
37	D	3	91
38	B	6	184
39	A	4	104
40	C	2	55

Licensed For Use Only By: Jose Del Castillo j.delcastillos@gmail.com Nov 30 2019 12: